STPM 2014 PHYSICS 960/1

Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper. The retyped paper does not show part marks; the marks given here are a suggested allocation.

Section A [15 marks]

1. What is the dimension of a derived quantity, $\frac{1}{2}\rho v^2$, where ρ is the density of a fluid and v is its speed?

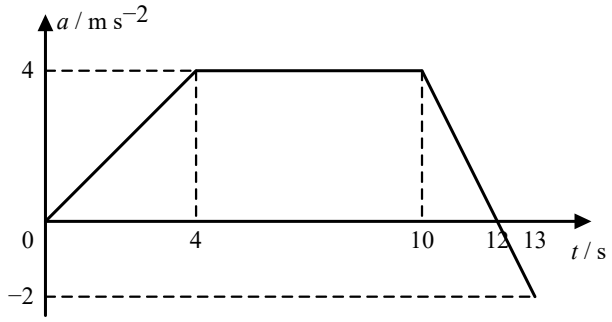
A $ML^{-1}T^{-1}$ B $ML^{-1}T^{-2}$ C ML^2T^{-1} D ML^2T^{-2}

Answer: B

$$[\tfrac{1}{2}\rho v^2] = (ML^{-3})(LT^{-1})^2 = ML^{-1}T^{-2}$$

Explanation: The factor $\frac{1}{2}$ has no dimensions. The result has the dimensions of pressure, as expected for the dynamic pressure of a fluid.

2. The variation of acceleration a with time t for a car which is initially at rest is shown in the graph below.



The maximum velocity that can be reached by the car is

A $32.0\ m\ s^{-1}$ B $35.0\ m\ s^{-1}$ C $36.0\ m\ s^{-1}$ D $37.0\ m\ s^{-1}$

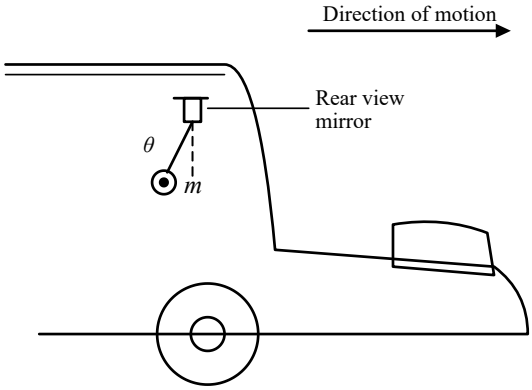
Answer: C

The velocity is greatest when the acceleration falls to zero, at $t = 12$ s. Change in velocity = area under the a–t graph:

$$\tfrac{1}{2}(4)(4) + (6)(4) + \tfrac{1}{2}(2)(4) = 8 + 24 + 4 = 36.0\ m\ s^{-1}$$

Explanation: After 12 s the acceleration is negative, so the car starts to slow down. Do not include the area below the axis.

3. A decorative CD of mass m hanging from the rear view mirror of a car which is accelerating is shown in the diagram below.



If the tension in the string is T and the acceleration of the car is a, which equation about the motion of the CD is correct?

A $T\cos\theta = ma$ B $T\sin\theta = ma$ C $\dfrac{T}{\cos\theta} = ma$ D $\dfrac{T}{\sin\theta} = ma$

Answer: A

With θ measured between the string and the horizontal, the horizontal component of the tension, $T\cos\theta$, gives the CD the same acceleration as the car:

$$T\cos\theta = ma$$

Explanation: Vertically, $T\sin\theta = mg$. Only the horizontal component accelerates the CD forward with the car. (If the angle were measured from the vertical, the components would swap.)

4. A passenger of mass 50 kg is in a car which is moving with a velocity of 30 m s^{-1}. If the driver suddenly brakes the car, the passenger who is wearing a seat belt is pushed forward a distance of 0.5 m. What is the average pulling force of the seat belt and the time interval of the force acted on the passenger?

[Neglect the effect of friction between the passenger and the seat of the car.]

	Average force	*Time interval*
A	4.9×10^2 N	0.03 s
B	4.9×10^2 N	3.06 s
C	4.5×10^4 N	0.03 s
D	4.5×10^4 N	3.06 s

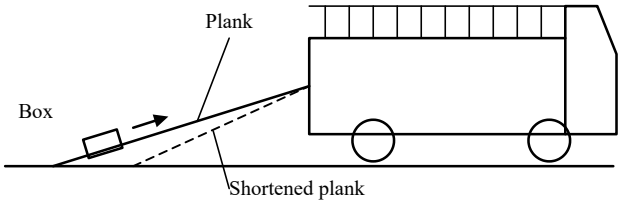
Answer: C

$$a = \frac{u^2}{2s} = \frac{30^2}{2(0.5)} = 900 \text{ m s}^{-2} \Longrightarrow F = (50)(900) = 4.5 \times 10^4 \text{ N}$$

$$t = \frac{u}{a} = \frac{30}{900} = 0.03 \text{ s}$$

Explanation: The passenger stops from 30 m s^{-1} over 0.5 m. The large force acting for a short time shows why seat belts are designed to stretch slightly.

5. An inclined plank which is used to transfer a box from the ground up onto a lorry is shown in the diagram below.



If the plank is shortened, which statement is true of the work done to transfer the same box?

A The work done against friction decreases.
B The work done against friction increases.
C The work done against gravity decreases.
D The work done against gravity increases.

Answer: A

Work against friction = $\mu mg \cos\theta \times L = \mu mg \times$ (horizontal distance). A shorter plank covers a smaller horizontal distance, so the work done against friction decreases.

Explanation: The work done against gravity, mgh, depends only on the height of the lorry, so it is unchanged.

6. A mass tied to one end of a string is whirled in a horizontal circle with uniform speed. Which statement is true of the motion of the mass?

A An outward force acts on the mass.
B The net force acting on the mass is zero.
C The tension in the string is balanced by an outward force on the mass.
D The tension in the string enables the mass to move in a circular path.

Answer: D

The tension in the string provides the centripetal force that keeps the mass moving in a circle.

Explanation: There is no real outward force on the mass, and the net force is not zero: it is the centripetal force directed towards the centre.

7. The mass of the Earth is 80 times the mass of the Moon. The distance of the Moon from the Earth is 3.8×10^8 m. The distance from the Earth of a satellite with resultant gravitational field strength zero is

A 1.90×10^8 m B 3.42×10^8 m C 3.75×10^8 m D 3.80×10^8 m

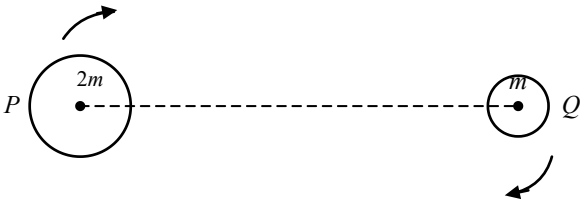
Answer: B

$$\frac{GM}{x^2} = \frac{G(M/80)}{(d-x)^2} \Rightarrow \frac{d-x}{x} = \frac{1}{\sqrt{80}}$$

$$x = \frac{3.8 \times 10^8}{1 + 1/\sqrt{80}} = 3.42 \times 10^8 \text{ m}$$

Explanation: At the neutral point the fields of the Earth and Moon are equal and opposite. It is much closer to the Moon because the Moon's mass is smaller.

8. Two stars P and Q of mass $2m$ and m respectively, which rotate about the centre of mass of the system, are shown in the diagram below.



What is the ratio of speed of P to speed of Q?

A $\frac{1}{4}$ B $\frac{1}{3}$ C $\frac{1}{2}$ D $\frac{2}{3}$

Answer: C

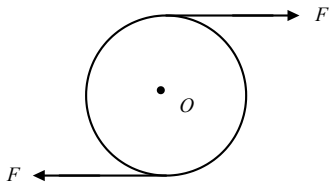
Centre of mass: $(2m)r_P = mr_Q \Rightarrow r_P : r_Q = 1 : 2$

Same angular velocity: $\frac{v_P}{v_Q} = \frac{r_P\omega}{r_Q\omega} = \frac{1}{2}$

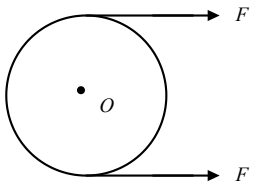
Explanation: Both stars complete one revolution in the same time, so their speeds are proportional to their distances from the centre of mass.

9. A pair of forces F of equal magnitude acts on a solid disc which is free to rotate about an axis through O. In which configuration is the disc in static equilibrium?

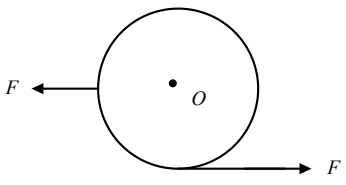
A



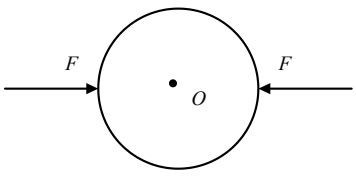
B



C



D

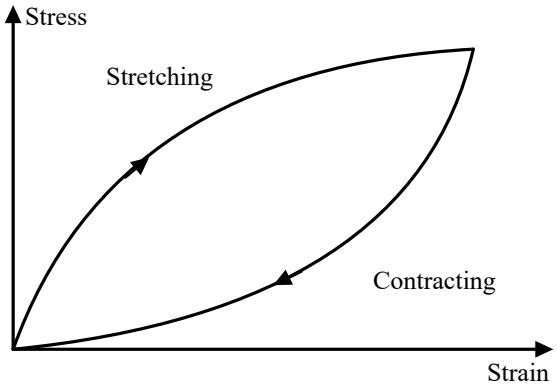


Answer: D

In D the two equal and opposite forces act along the same line through O: the resultant force and the resultant torque are both zero.

Explanation: A is a couple (zero resultant force but a turning effect); B has a resultant force of $2F$; C has both a resultant force and a torque.

10. A stress-strain curve for a material is shown in the graph below.



The material was first stretched, and then allowed to contract. Which deduction is true of the material?

A The material was stretched beyond the elastic limit.
B The material was permanently deformed after the process.
C Net work done in the process is stored as strain energy in the material.
D More work done is involved in stretching than in contracting the material.

Answer: D

The stretching curve lies above the contracting curve, so the area under it (work done in stretching) is greater than the work given back during contraction.

Explanation: The material returns to zero strain, so it is not permanently deformed. The area of the loop is energy dissipated as heat, not stored as strain energy.

11. Which statement about the law of equipartition of energy is true?

A The energy of a molecule is equally divided among all degrees of freedom where each degree of freedom has an average energy of $\frac{1}{2}kT$.
B All gas molecules have the same number of degrees of freedom where each degree of freedom has an average energy of $\frac{1}{2}kT$.
C All gas molecules have three degrees of freedom for translational motion where each molecule has an average energy of $\frac{1}{2}kT$.
D The energy of an ideal gas is equally divided among all the molecules where each molecule has an average energy of $\frac{1}{2}kT$.

Answer: A

Equipartition: the energy of a molecule is shared equally among all its degrees of freedom, each having an average energy of $\frac{1}{2}kT$.

Explanation: Different molecules have different numbers of degrees of freedom (B wrong). Each **degree of freedom**, not each molecule, has $\frac{1}{2}kT$.

12. At a fixed temperature, what is the value of the ratio $\frac{\text{r.m.s. speed of oxygen gas molecule}}{\text{r.m.s. speed of carbon dioxide gas molecule}}$?

[Relative atomic masses of carbon and oxygen are 12 and 16 respectively.]

A 0.85 B 1.17 C 1.38 D 3.46

Answer: B

$$\frac{c_{O_2}}{c_{CO_2}} = \sqrt{\left(\frac{M_{CO_2}}{M_{O_2}}\right)} = \sqrt{\left(\frac{44}{32}\right)} = 1.17$$

Explanation: At the same temperature, the r.m.s. speed is inversely proportional to the square root of the molar mass ($O_2 = 32$, $CO_2 = 12 + 2 \times 16 = 44$).

13. The molar heat capacity of a monatomic ideal gas at constant volume is

A $\frac{R}{2}$ B R C $\frac{3}{2}R$ D $\frac{5}{2}R$

Answer: C

$$C_{V,m} = \frac{f}{2}R = \frac{3}{2}R$$

Explanation: A monatomic gas has 3 translational degrees of freedom, each contributing $\frac{1}{2}R$.

14. An ideal gas undergoes the following thermodynamic processes:

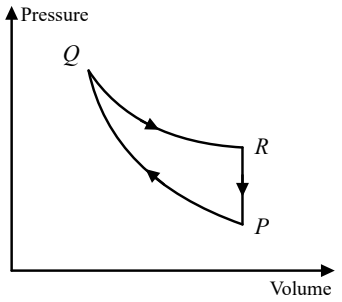
$P \rightarrow Q$: Adiabatic compression to half of its initial volume.

$Q \rightarrow R$: Isothermal expansion back to its initial volume.

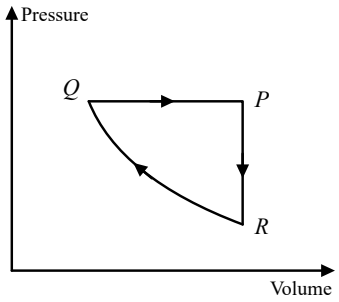
$R \rightarrow P$: Cooling down at a constant volume back to its initial state.

Which graph represents the above processes?

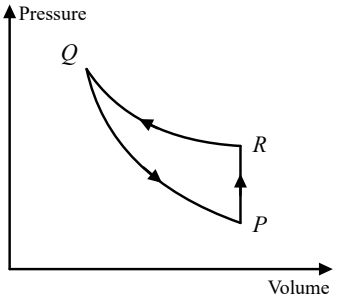
A



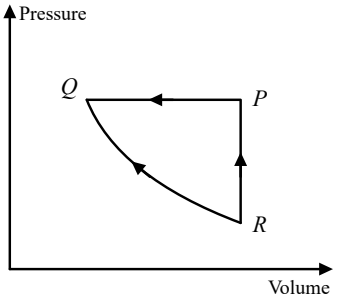
B



C



D



Answer: A

$P \rightarrow Q$: adiabatic compression, a steep curve to smaller volume and higher pressure.

$Q \rightarrow R$: isothermal expansion, a less steep curve back to the original volume.

$R \rightarrow P$: constant volume cooling, vertical line downward. This matches A.

Explanation: An adiabatic curve is steeper than an isothermal curve, so the adiabatic and isothermal lines form a narrow loop closed by the vertical line.

15. A refrigerator door is 150.0 cm high, 80.0 cm wide and 6.0 cm thick. The temperatures of the inner and outer surfaces of the door are 0 °C and 30 °C respectively. What is the heat flow through the door in one minute?

[The thermal conductivity of the door is 0.21 W m^{-1} K^{-1}.]

A 1.26×10^2 J B 7.56×10^2 J C 1.27×10^2 J D 7.64×10^2 J

Answer: B

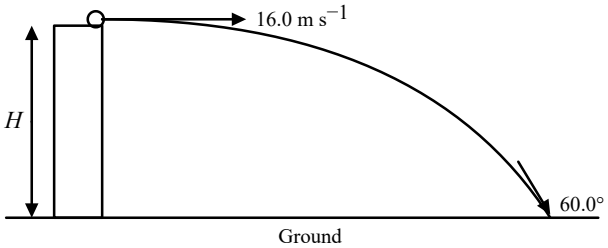
$$Q = \frac{kA\Delta\theta t}{x} = \frac{(0.21)(1.50 \times 0.800)(30)(60)}{0.060} = 7.56 \times 10^3 \text{ J}$$

This corresponds to option B (printed as 7.56×10^2 J).

Explanation: The area is the height × width of the door, and the thickness is the distance the heat travels. Remember to multiply the rate by 60 s.

Section B [15 marks]

16. A ball is thrown horizontally from the roof of a building of height H as shown in the diagram below.



The initial speed of the ball is 16.0 m s^{-1}. It hits the ground at an angle of 60.0° with the horizontal.

(a) Assuming that air resistance is negligible, determine

(i) the vertical component of the velocity of the ball when it hits the ground, [3]

Answer

Horizontal velocity stays 16.0 m s^{-1} — 1

$\tan 60.0° = \frac{v_y}{16.0}$ — 1

$v_y = 27.7$ m s^{-1} — 1

3

Explanation: The angle at which the ball hits the ground is the direction of its velocity, given by the ratio of the vertical to the horizontal components.

(ii) the height H of the building. [2]

Answer

$v_y^2 = 2gH \Longrightarrow H = \frac{27.7^2}{2(9.81)}$ — 1

$H = 39.1$ m — 1

2

Explanation: Vertically the ball starts with zero velocity and falls freely.

(b) If there is air resistance, what is the effect on the angle when the ball hits the ground? Give a reason for your answer. [2]

Answer

The angle becomes larger (the ball hits the ground more steeply) — 1

because air resistance reduces the horizontal component of velocity throughout the flight, while the vertical motion is mainly controlled by gravity — 1

2

Total for question **7**

Explanation: With air resistance the ball travels a shorter horizontal distance, and the ratio v_y/v_x at impact is greater.

17. The molar heat capacity of a certain ideal gas at constant volume is 24.95 J K^{-1} mol^{-1}. A cylinder containing 2.0 mol of the gas at 320 K is heated at constant volume to 350 K. Determine

(a) the amount of heat required to heat the gas, [2]

Answer

$Q = nC_{V,m}\Delta T = (2.0)(24.95)(350 - 320)$ 1

$Q = 1.50 \times 10^3$ J (1497 J) 1

2

Explanation: At constant volume no work is done, so all the heat raises the internal energy.

(b) the molar heat capacity of the gas at constant pressure, [2]

Answer

$C_{p,m} = C_{V,m} + R = 24.95 + 8.31$ 1

$C_{p,m} = 33.26$ J K^{-1} mol^{-1} 1

2

Explanation: For an ideal gas the two molar heat capacities differ by the molar gas constant.

(c) the ratio of molar heat capacity at constant pressure to molar heat capacity at constant volume of the gas, [2]

Answer

$\gamma = \frac{C_{p,m}}{C_{V,m}} = \frac{33.26}{24.95}$ 1

$\gamma = 1.33$ 1

2

Explanation: A value of $1.33 = \frac{4}{3}$ is typical of a polyatomic (non-linear) molecule.

(d) the degrees of freedom of the gas. [2]

Answer

$C_{V,m} = \frac{f}{2}R \Longrightarrow f = \frac{2(24.95)}{8.31}$ 1

$f = 6$ 1

2

Total for question **8**

Explanation: Six degrees of freedom: 3 translational + 3 rotational, as for a non-linear polyatomic molecule. (Equivalently, $\gamma = 1 + 2/f$.)

Section C [30 marks]

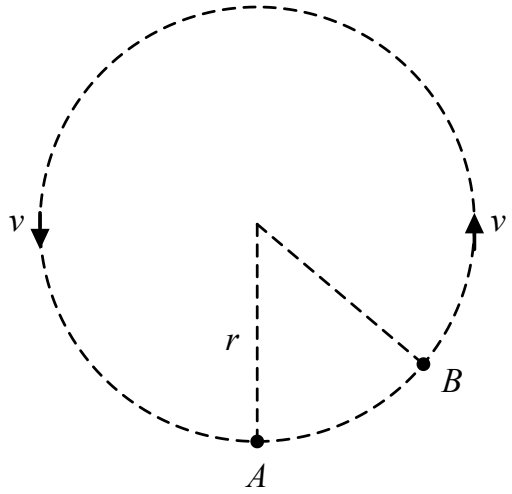
18. (a) Explain why a particle moving in a circular path with constant speed has acceleration. [2]

Answer

The direction of motion (velocity) changes continuously although the speed is constant	1
A change in velocity means there is an acceleration, directed towards the centre	1
	2

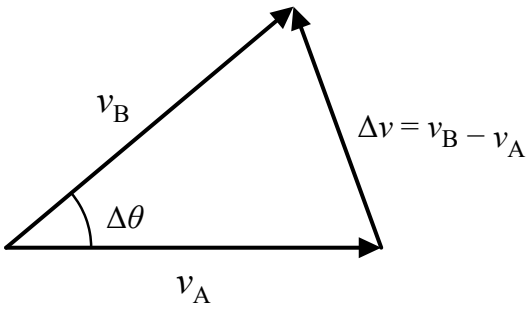
Explanation: Velocity is a vector, so a change in direction alone is a change in velocity.

(b) A particle moves from position A to position B in a circular path of radius r with constant speed v in time Δt as shown in the diagram below.



(i) Draw a vector diagram for the change in velocity Δv. [2]

Answer



$\boldsymbol{v}_A$ and $\boldsymbol{v}_B$ drawn from the same point, equal in length, at angle $\Delta\theta$	1
$\Delta\boldsymbol{v}$ from the tip of $\boldsymbol{v}_A$ to the tip of $\boldsymbol{v}_B$	1
	2

Explanation: $\Delta \boldsymbol{v} = \boldsymbol{v}_B - \boldsymbol{v}_A$; for a small angle it points towards the centre of the circle.

(ii) Show that the angle between the velocity vectors at A and B is $\frac{v\Delta t}{r}$. [2]

Answer

Arc length $AB = v\Delta t = r\Delta\theta \Longrightarrow \Delta\theta = \frac{v\Delta t}{r}$ — 1

Each velocity is perpendicular to its radius, so the angle between $\boldsymbol{v}_A$ and $\boldsymbol{v}_B$ equals the angle $\Delta\theta$ turned by the radius — 1

2

Explanation: Angles in radians: arc length = radius × angle.

(iii) Derive the expression for the centripetal acceleration a in terms of v and r. [2]

Answer

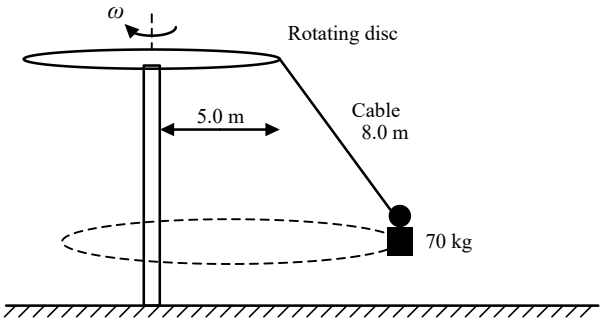
For small $\Delta\theta$: $|\Delta v| = v\Delta\theta = \frac{v^2\Delta t}{r}$ — 1

$a = \frac{|\Delta v|}{\Delta t} = \frac{v^2}{r}$ — 1

2

Explanation: The vector triangle is isosceles with two sides v; for a small angle the third side is $v\Delta\theta$.

(c) A large spinning swing revolving about a vertical axis is shown in the diagram below.

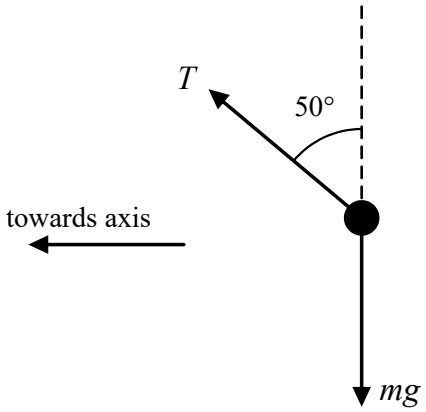


The swing with a cable length 8.0 m and a boy with a mass of 70.0 kg, including the chair he is sitting on, are swung from the edge of a rotating disc with a radius of 5.0 m. The boy moves in a

horizontal circular path with the cable making an angle of inclination 50° with the vertical axis. Neglect the air resistance and the mass of the cable.

(i) Sketch the forces which act on the boy. [2]

Answer



Tension T along the cable, at 50° to the vertical	1
Weight mg vertically downward	1
	2

Explanation: Only two forces act. Their resultant is horizontal, towards the axis: the centripetal force. There is no separate 'centrifugal' force.

(ii) Calculate the tension in the cable. [2]

Answer

Vertical: $T \cos 50° = mg = (70.0)(9.81)$	1
$T = 1.07 \times 10^3$ N	1
	2

Explanation: The boy does not move vertically, so the vertical component of the tension balances his weight.

(iii) Determine the speed of the boy. [3]

Answer

Radius of the circle: $r = 5.0 + 8.0 \sin 50° = 11.13$ m	1
Horizontal: $T \sin 50° = \frac{mv^2}{r} \Rightarrow v^2 = gr \tan 50°$	1
$v = 11.4$ m s^{-1}	1
	3
Total for question	**15**

Explanation: The boy's circle has radius equal to the disc radius plus the horizontal reach of the cable.

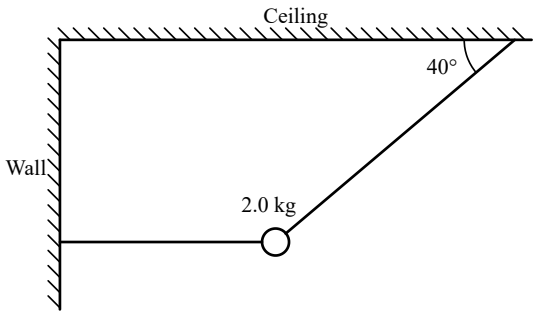
19. (a) State the conditions for the equilibrium of a rigid body. [2]

Answer

The resultant force acting on the body is zero	1
The resultant torque about any point is zero	1
	2

Explanation: Both conditions are needed for a rigid body to be in equilibrium.

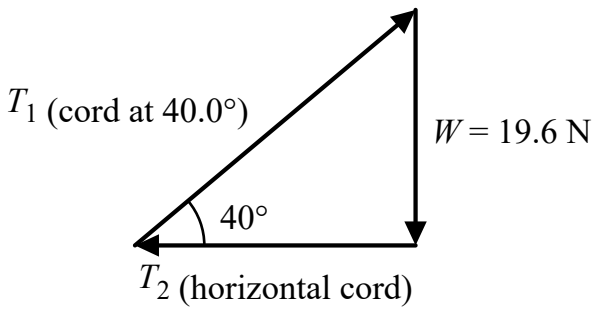
(b) A spherical steel ball of mass 2.0 kg is supported by two cords fixed to a ceiling and a wall as shown in the diagram below.



The angle between the cord and the ceiling is 40.0° while the other cord is attached horizontally to the steel ball.

(i) Sketch a triangle of forces representing the forces in equilibrium. [2]

Answer



Closed triangle, arrows head to tail	1
Forces labelled: weight, horizontal tension and tension at 40.0°	1
	2

Explanation: The three forces on the ball are in equilibrium, so they form a closed triangle when drawn to scale head to tail.

(ii) Determine the tensions in the two cords. [3]

Answer

Vertical: $T_1 \sin 40.0° = (2.0)(9.81) = 19.6$ N	1
$T_1 = 30.5$ N (cord to the ceiling)	1
Horizontal: $T_2 = T_1 \cos 40.0° = 23.4$ N (horizontal cord)	1
	3

Explanation: Only the inclined cord has a vertical component, so it alone supports the weight.

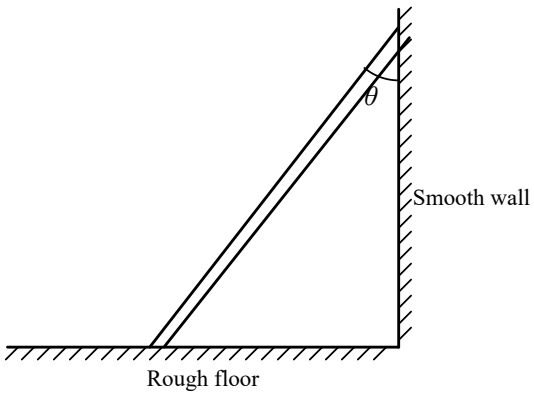
(iii) If the breaking tension of both cords is 45 N, calculate the maximum mass of the steel ball which can be supported by the cords. [2]

Answer

The inclined cord has the greater tension, so it breaks first: $45 \sin 40.0° = mg$	1
$m = 2.95$ kg	1
	2

Explanation: $T_1 > T_2$ always here, so set the larger tension equal to the breaking tension.

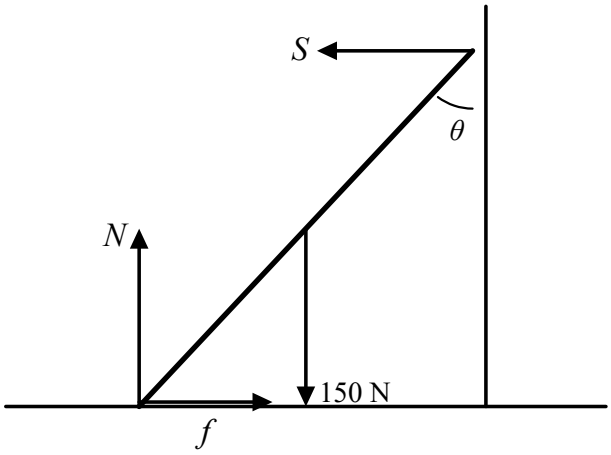
(c) A uniform ladder of weight 150 N leans against a smooth wall and is placed on a rough floor as shown in the diagram below.



The coefficient of friction between the floor and the ladder is 0.38.

(i) Sketch a diagram showing all the forces acting on the ladder. [2]

Answer



Weight at the midpoint, normal reaction and friction (towards the wall) at the floor	1
Horizontal reaction at the smooth wall	1
	2

Explanation: A smooth wall can exert only a force perpendicular to its surface (horizontal).

(ii) Determine the angle θ when the ladder is about to slide. [4]

Answer

Horizontal: $S = f = \mu N$; vertical: $N = 150$ N $\Longrightarrow S = (0.38)(150) = 57$ N	1
Moments about the foot (length L): $S(L\cos\theta) = 150(\frac{L}{2}\sin\theta)$	1
$\tan\theta = \frac{2(57)}{150} = 0.76$	1
$\theta = 37°$	1
	4
Total for question	**15**

Explanation: θ is the angle between the ladder and the wall. The ladder is about to slide when friction at the floor reaches its limiting value μN.

20. (a) Describe the mechanism of heat conduction in a non-metallic solid. [2]

Answer

Atoms at the hotter end vibrate with greater amplitude (more kinetic energy) 1

They pass energy to neighbouring atoms through the bonds/collisions, so energy is transferred along the solid (lattice vibrations) 1

2

Explanation: A non-metal has no free electrons, so conduction is only by lattice vibrations, which is slow.

(b)

(i) Define the *thermal conductivity* of a material. [1]

Answer

Thermal conductivity is the rate of heat flow per unit cross-sectional area per unit temperature gradient 1

1

Explanation: $k = \frac{\mathrm{d}Q/\mathrm{d}t}{A(\mathrm{d}\theta/\mathrm{d}x)}$, unit W m^{-1} K^{-1}.

(ii) In determining the thermal conductivity of a poor heat conductor, the major problem is poor heat flow. Explain how it can be overcome. [2]

Answer

Use a thin specimen with a large cross-sectional area (a thin disc) 1

so that the rate of heat flow ($\propto A/x$) is large enough to measure 1

2

Explanation: This is the principle of Lees' disc method for poor conductors.

(c) The wall of a house is made of bricks with a thickness of 20.0 cm and has an area of 30.0 m^2. In the afternoon, the outside temperature of the house can reach 38.0 °C while the inside temperature is maintained at 18.0 °C. In order to reduce heat loss, plaster with a thickness of 15.0 mm is applied on the inside surface of the wall of the house. Assuming that the outside and inside temperatures of the house are constant, calculate

(i) the rate of heat flow through the wall without plaster, [2]

Answer

$\frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{kA\Delta\theta}{x} = \frac{(0.84)(30.0)(38.0 - 18.0)}{0.200}$ 1

$= 2.52 \times 10^3$ W 1

2

Explanation: Convert the thickness to metres.

(ii) the rate of heat flow through the wall with plaster, [3]

Answer

$\frac{dQ}{dt} = \frac{A\Delta\theta}{\Sigma(x/k)}$ 1

$= \frac{(30.0)(20.0)}{\frac{0.200}{0.84} + \frac{0.0150}{4.0 \times 10^{-3}}} = \frac{600}{3.988}$ 1

$= 150$ W 1

3

Explanation: The brick and plaster are in series; the thin plaster layer has a much larger thermal resistance (3.75) than the brick (0.238).

(iii) the interface temperature between the wall and the plaster, [3]

Answer

Same rate in each layer: $(0.84)(30.0)\frac{38.0 - \theta}{0.200} = 150$ 1

$38.0 - \theta = 1.19$ 1

$\theta = 36.8$ °C 1

3

Explanation: Most of the temperature drop is across the plaster, so the brick–plaster interface is close to the outside temperature.

(iv) the percentage of energy blocked by adding the plaster. [2]

Answer

Percentage blocked $= \frac{2520 - 150}{2520} \times 100\%$ 1

$= 94\%$ 1

2

Total for question **15**

Explanation: The plaster reduces the heat flow to about 6% of its value without plaster.

[The thermal conductivities of brick and plaster are 0.84 W m^{-1} K^{-1} and 4.0×10^{-3} W m^{-1} K^{-1}.]